PT ETANA BIOTECHNOLOGIES INDONESIA	STANDARD OPERATING PROCEDURE	Page : 1 of 18
	TITLE : **PERAWATAN TERHADAP MESIN FILLING (TOFFLON)** ***PREVENTIVE MAINTENANCE FILLING MACHINE (TOFFLON)***	SOP No. : SOP/EBI/EN-044 Revision : 01 Effective Date :

PREPARED BY :	PREPARED BY :	REVIEWED BY :	PREPARED BY :
PRIBADI SETIA BAYU AJI (NON UTILITY ENGINEERING TECHNICIAN)	**WENDI RUKMANSYAH (ENGINEERING SPV)**	**PURNO BUDI KISWANTO (ENGINEERING SR. MANAGER)**	**MUHAMAD WILDAN NUGRAHA (PROD SR. SPV)**

REVIEWED BY :	APPROVED BY :	APPROVED BY :
VINO SOADUAN (PROD SR. MANAGER)	**DEVIKA SRI WULANDARI (QS MANAGER)**	**SHANDY (QA MANAGER)**

1. TUJUAN

PURPOSE

Memberikan petunjuk tentang perawatan rutin terhadap mesin *Filling* (*Tofflon*) agar sesuai dengan standar yang ada di PT Etana Biotechnologies Indonesia.

Provides a guidance for maintenance of the Filling machine (Tofflon) in accordance with standards at PT. Etana Biotechnologies Indonesia.

2. RUANG LINGKUP

SCOPE

SOP ini digunakan oleh Departemen Engineering khususnya untuk teknisi sebagai pedoman dalam melakukan perawatan terhadap Mesin *Filling (Tofflon)* yang ada di area produksi lantai 1.

This SOP is used by engineering department especially for technician as guidance in order to maintain of Filling machine (Tofflon) in the production 1st floor.

3. TANGGUNG JAWAB

RESPONSIBILITY

3.1. Teknisi Engineering *Preventive Maintenance* bertanggung jawab:

Engineering Technician Preventive Maintenance are responsible:

3.1.1. Melakukan tugas perawatan dengan baik, sesuai dengan SOP ini.

Performs maintenance activities properly, in accordance with this SOP.

3.1.2. Melakukan perawatan mesin *Filling (Tofflon)* sesuai jadwal yang telat ditetapkan.

Performs maintenance of Filling machine (Tofflon) according to a predefined schedule.

3.1.3. Melaporkan setiap terjadi kerusakan dan perbaikan yang diperlukan kepada Supervisor Engineering.

Report any damage and repairs that are needed to the engineering supervisor.

3.2. Supervisor Engineering bertanggung jawab:

Engineering Supervisor are responsible:

3.2.1. Membuat jadwal perawatan mesin *Filling* (*Tofflon)* secara berkala.

Make schedule of filling machine (Tofflon) maintenance periodically.

3.2.2. Memastikan pelaksanaan perawatan sesuai dengan jadwal.

Make sure the implementation of maintenance is according to the schedule.

3.2.3. Memastikan semua alat ukur yang terlibat dalam proses perawatan sesuai dengan spesifikasi yang telah ditentukan (dikalibrasi).

Make sure all measuring instruments involved in the maintenance process are in accordance with the specifications specified (calibrated).

3.2.4. Mengadakan pemeriksaan bulanan terhadap hasil kerja teknisi *Preventive Maintenance*.

Perform a monthly review of the workings of the Preventive Maintenance Technicians.

3.2.5. Memberikan pelatihan kepada departemen lain yang terkait.

Provide training to others department which related.

3.3. Manajer Engineering bertanggung jawab:

Engineering Manager is responsible:

3.3.1. Menentukan *spare parts* pengganti apabila *spare parts* yang sesuai sudah tidak dijual lagi.

Determine replace parts if suitable spare part are no longer sold.

3.3.2. Memutuskan dan mengkoordinir tindakan yang diperlukan, apabila perlu dilakukannya penggantian, modifikasi peralatan, perubahan sistem dan menyiapkan prosedur perubahan.

Resolve and coordinate necessary actions, if necessary replacement, equipment modification, system changes and prepared change control.

4. DEFINISI

DEFINITION

4.1. PM (*Preventive Maintenance*) adalah perawatan rutin yang terjadwal untuk memastikan peralatan atau aset tetap berfungsi normal sesuai dengan fungsinya dan untuk mencegah kerusakan yang tidak terduga di masa mendatang.

PM (Preventive Maintenance) is scheduled routine maintenance to ensure equipment or assets continue to function normally according to their function and to prevent unexpected damage in the future.

4.2. Mesin pengisian otomatis adalah mesin yang digunakan untuk membantu mengisi produk dalam bentuk cairan, pengisian produk cair dapat dilakukan dalam kemasan primer seperti botol, vial, ampul, maupun *prefilled syringe*. Penggunaan mesin *filling* cairan otomatis membuat volume dalam kemasan primer lebih akurat sesuai dengan volume yang sudah ditentukan.

Automatic filling machines are machines used to help fill products in liquid form, filling liquid products can be done in primary packaging such as bottles, vials, ampoules, or prefilled syringes. The use of automatic liquid filling machines makes the volume in the primary packaging more accurate in accordance with the specified volume.

4.3. *Stoppering* merupakan proses pemasangan *rubber stopper* ke *syringe* setelah proses *dosing.*

Stoppering is the process of attaching the rubber stopper to the syringe after the dosing process.

4.4. *P&P* merupakan singkatan dari *Pick and Place*, adalah bagian dari mesin berbentuk *silinder* untuk memindahkan *nest* dari *conveyor* ke *tray.*

P&P is abbreviation of pick and place, this part like hand for moving the nest from conveyor to tray.

4.5. Pompa peristaltik merupakan salah satu jenis pompa *positive displacement*. Pompa peristaltik dapat digunakan untuk memompakan berbagai macam cairan.

Peristaltic Pump is Peristaltic pump is one types of pumps with positive displacement. *Peristaltic pumps can be used to pump various liquids.*

4.6. *WFI (Water For Injection)* adalah air murni yang bebas p*rogen. WFI* dapat diperoleh dengan cara mengolah air murni biasa (*Feed Water*) secara penyulingan

WFI (Water For Injection) is pyrogen-free, pure water. WFI can be obtained by processing ordinary pure water (feed water) by distillation.

4.7. *Autonomous Maintenance* adalah perawatan mandiri mesin yang dilakukan oleh operator mesin. Bila selama ini operator hanya dilatih untuk mengoperasikan mesin, maka sudah saatnya untuk dilatih lebih lanjut.

Autonomous maintenance is an independent engine maintenance carried out by the machine operator. If all this time the operators are only trained to operate the engine, then it's time to be further trained.

5. PETUNJUK UMUM

GENERAL INSTRUCTION

5.1. Cara perawatan harus dilakukan dengan teratur dan sesuai dengan jadwal.

The way of maintenance should be done regularly and in accordance with the schedule.

5.2. Teknisi yang telah terlatih atau sudah mendapatkan pelatihan dan terkualifikasi yang dapat melakukan perawatan sistem mesin *Filling Tofflon*

Trained or qualified training technicians who can conduct maintenance of Filling machine Tofflon system.

5.3. Keselamatan kerja harus tetap diperhatikan selama perawatan.

Work safety must be kept in mind during the maintenance.

5.4. Kegiatan perawatan yang dilakukan secara berkala harus didokumentasikan setelah perawatan dilakukan.

Regular maintenance activities should be documented after maintenance done.

5.5. Apabila alat ukur tersebut didapati dalam keadaan rusak, harus diberi penandaan yang jelas.

If the measuring instrument is found to be in a defective state, it should be clearly marked.

5.6. Perlengkapan kerja harus sesuai dengan yang telah ditetapkan.

Work equipment must be in accordance with the set.

5.7. Penggantian *spare part* harus sesuai dengan spesifikasi dan fungsinya.

Spare parts replacement must be in accordance with the specifications and functions.

5.8. Untuk perubahan major yang mendasar dan merubah sistem *mesin Filling Tofflon* sistem harus memakai formulir pengendalian perubahan dan disetujui oleh manajer terkait.

For major underlying changes and changing the Filling machine Tofflon system must use a change control form and be approved by related managers.

6. ALAT DAN BAHAN

TOOLS AND MATERIALS

6.1. ALAT

TOOLS

6.1.1. satu set kunci pas

A set of wrench

6.1.2. Satu set sarung obeng

A set of screwdriver

6.1.3. Satu set kunci L.

A set of allen key

6.1.4. Multimeter

Mulitimeter

6.2. BAHAN

MATERIAL

6.2.1. Alkohol 70%

Alcohol 70%

6.2.2. Kain Majun

Wipes fabric

6.2.3. *Grease*

Grease

7. PROSEDUR

PROCEDURE

Perawatan berkala pada Mesin *Filling Tofflon*.

Maintenance of filling tofflon machine.

7.1. Perawatan berkala setiap 1 bulanan (L1) pada mesin *Filling Tofflon*

Regular maintenance every 1 month (L1) on Tofflon Filling machine

7.1.1. Perawatan mekanikal dan elektrikal.

Mechanical and electrical maintenance.

Tabel 1. Perawatan Rutin Mesin *Filling*
Table 1. Preventive Maintenance of Filling Machine

Perawatan *Maintenance*	Jangka Waktu *Period*
Pelumasan Mesin *Filling* *Lubrication of the Filling Machine*	1 bulan / *1 month*
Pemeriksaan Sensor dan Detektor *Checking Sensors and Detectors*	1 bulan / *1 month*
Pemeriksaan Roda Gigi *Checking Gear*	1 bulan / *1 month*
Pemeriksaan pada Pompa Peristaltik *Checking the Peristaltic Pump*	1 bulan / *1 month*
Pemeriksaan kondisi *Stoppering rubber* *Checking Stoppering Rubber Condition*	1 bulan / *1 month*
Pemeriksaan *Belt Conveyor* *Checking Belt Conveyor*	1 bulan / *1 month*
Pemeriksaan Koneksi pada Panel Elektrikal *Checking Electrical Panel for Wiring and Connection*	1 bulan / *1 month*
Pemeriksaan Pompa Vakum Checking the Vacuum Pump	1 bulan / *1 month*
Penggantian Oli Pompa Vakum *Changing the Vacuum Pump Oil*	1 bulan / *1 month*
Pemeriksaan Filter Pompa Vakum *Changing the Vacuum Pump Filter*	1 bulan / *1 month*
Pemeriksaan Kondisi *Vibrator Bowl* *Checking Vibrator Bowl Condition*	1 bulan / *1 month*
Pemeriksaan *Shaft Stoppering* *Checking of Shaft Stoppering*	1 bulan / *1 month*
Pemeriksaan pada Linear Track *Checking the Linear Track*	1 bulan / *1 month*

Pengecekan bagian Mesin *Filling* sebelum dan sesudah melakukan proses, mesin *Filling* perlu dicek terlebih dahulu untuk memastikan semua bagian yang digunakan berfungsi normal. Dapat dilihat pada tabel 2.

Checking Filling Machine Parts before and after carrying out the process, the Filling machine needs to be checked first to ensure that all parts used are functioning normally. Can be seen in table 2.

Tabel 2. Pengecekan Mesin Filling

Table 2. Checking the Filling Machine

Deskripsi *Description*	Jangka Waktu *Period*
Cek Fungsi Pompa Vakum *Check Vacuum Pump Function*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Kondisi Oli Pompa Vakum *Check Vacuum Pump Oil Condition*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Kondisi *Solenoid* Vakum *Check Solenoid Vacuum Condition*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Fungsi Sensor Vakum *Check Vacuum Sensor Function*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Kondisi Filter Vakum *Check Vacuum Filter Condition*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Fungsi Servo *Stoppering Station Check Servo Stoppering Station Function*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek *Rubber Silicon Stoppering Check Rubber Silicon Stoppering*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Kondisi Konveyor *Check Conveyor Condition*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Kondisi *Filing Needle Check Filling Needle Condition*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Kondisi *Magazzine Bolt Check Magazzine Bolt Condition*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Kondisi *Magazzine Check Magazzine Condition*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Fungsi *Vibration Magazzine Check Vibration Magazzine Function*	Sebelum Proses dan Setelah Proses *Before Process and After Process*
Cek Kondisi & Fungsi *Regulator Vacuum Robot Arm Check the Condition & Function of the Robot Arm's Vacuum Regulator*	Sebelum Proses dan Setelah Proses *Before Process and After Process*

7.1.2. Pelumasan Manual

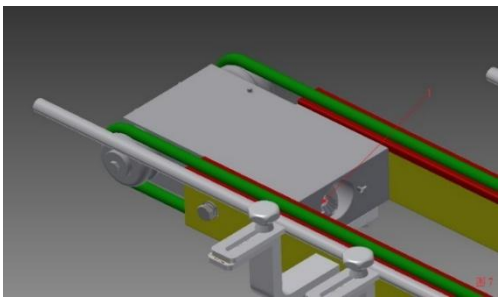
Point of the Lubrication

Mesin ini memiliki beberapa titik pelumasan *manual.* Pelumasan dilakukan di area *Conveyor belt main drive bevel gear*. Dapat dilihat pada gambar 1 & 2 *Manual Lubrication.*

This machine has some point of lubrication. Do lubrication in Conveyor belt main drive bevel gear area. Can be seen in figure 1 & 2.

Gambar 1 . Titik Pelumasan

Figure 1. Point of the Lubrication



Gambar 2. Area *Conveyor belt main drive bevel gear*

Figure 2. Conveyor belt main drive bevel gear area

7.1.2.1. Tipe *Grease* yang digunakan untuk pelumasan bagian.

Grease type used for part lubrication.

Grease yang digunakan harus aman untuk makanan disarankan menggunakan *Grease Wurth Muti-Purpose III.* Dapat dilihat pada gambar 3.

The grease must be food safe, commend to use the Wurth Muti-Purpose Grease III.

Gambar 3. Pelumas

Figure 3. Grease

7.1.3. Pemeriksaan *Sensor dan Detektor*

Sensors and Detectors

Periksa fungsi dari *sensor* dan *detektor* beroperasi dengan baik, perbaiki jika ada kerusakan atau ganti dengan yang baru.

Checking the function of sensors and detectors, ensuring works fine. Changing if broken or not work normally

Gambar 4. Sensor

Figure 4. Sensors

Gambar 5. Detektor

Figure 5. Detectors

7.1.4. Pemeriksaan Roda Gigi

Gear Inspection

Buka panel bawah mesin *Filling Tofflon.* Pemeriksaan visual pada setiap *gear*.

Open the panel under the Filling Machine Tofflon. Visual inspection all of the gear.

7.1.4.1 Pastikan tidak ada bagian yang retak.

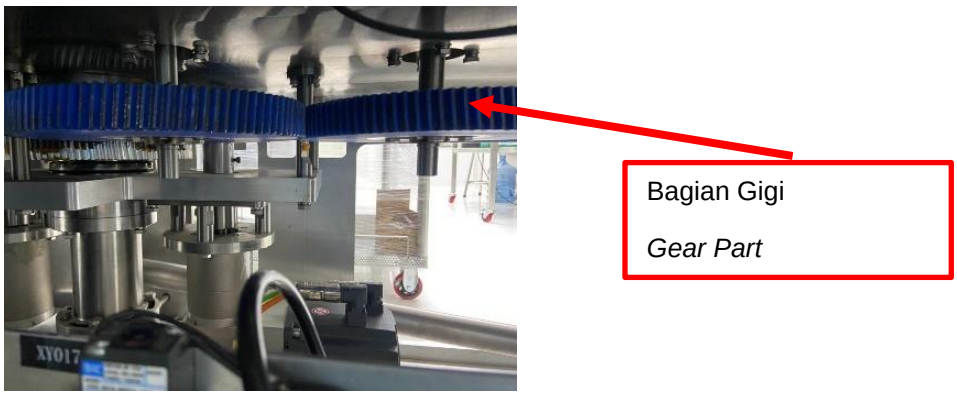
Make sure the part is no crack.

7.1.4.2 Pastikan di roda gigi tidak ada bagian antar gigi atau *Pitch* yang patah.

Make sure in the gear, there is no part between gears and pitch that broken.

7.1.4.3 Ganti *gear* yang rusak jika diperlukan.

Changes the gear if needed.



Gambar 6. Roda Gigi

Figure 6. Gear

7.1.5. Pemeriksaan Pompa Peristaltik

Peristaltic Pump

Pengecekan pada *peristaltic pump.* Pengecekan visual dan fungsi pada *Peristaltic pump filling tofflon.*

Checking the peristaltic pump. Visual inspection and function test of the peristaltic pump.

7.1.5.1 Periksa kondisi *Peristaltic Pump Cover*

Check the peristaltic pump cover condition.

Gambar 7. *Peristaltic Pump Cover*

Figure 7. Peristaltic Pump Cover

7.1.5.2 Periksa kondisi *Pump Peristaltic Roller.*

Check the peristaltic pump roller condition.

Gambar 8. *Peristaltic Pump Roller*

Figure 8. Peristaltic Pump Roller

7.1.6. *Pemeriksaan Stoppering Rubber*

Stoppering Rubber inspection

Pastikan kondisi *stoppering rubber* untuk memastikan *rubber* yang masuk ke *syringe* atau *vial* pas. Cara untuk mengatur *center* ialah dengan memastikan baut kencang untuk mencegah terjadinya masalah.

Make sure the condition of the stopper rubber is good to ensure that the rubber that enters the syringe or vial fits properly. The way to adjust the center is to make sure the bolts are tight to prevent problems

Gambar 9. *Stoppering Rubber*

Figure 9. Stoppering Rubber

7.1.7. Konveyor

Conveyor

Periksa konveyor, seperti :

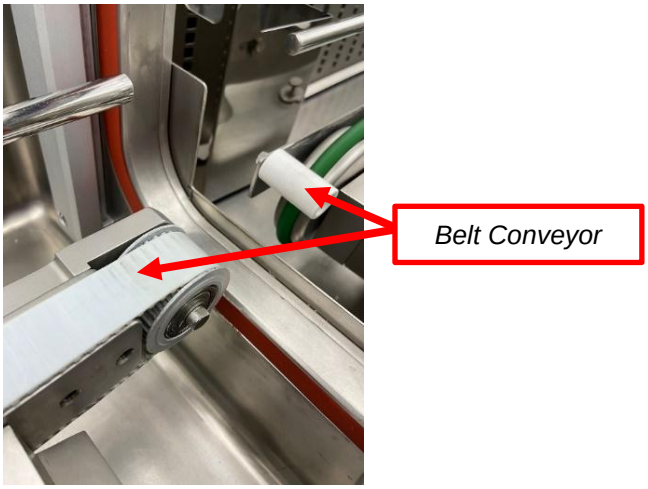
Checking the conveyor, such as :

7.1.9.1 Periksa sambungan konveyor.

Check the connection of conveyor

7.1.9.2 Periksa pinggiran *belt,* tidak rusak seperti terkelupas serat-seratnya.

Check the edge of the belt; it is not damaged as if the fibers have been peeled off



Gambar 10 Conveyor

Figure 10 Conveyor

7.1.8. Pemeriksaan koneksi pada Panel Elektrikal

Checking Electrical Panel for Wiring and Connection

7.1.8.1 Periksa kabel *power*, *ground* dan *netral* terpasang kuat dengan baut.

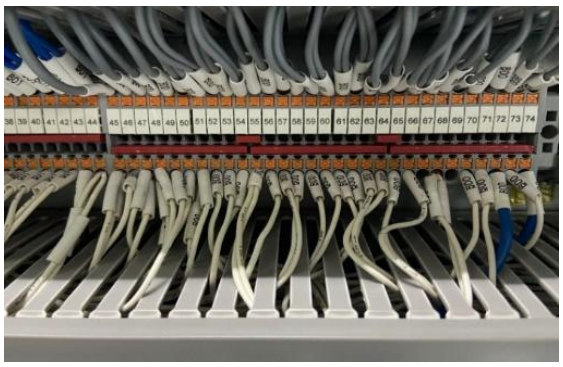
Check that the power, ground, and neutral cables are securely attached with screws.

7.1.8.2 Periksa *emergency shut down* bekerja dengan baik.

Checking the emergency shut down is work normally.

7.1.8.3 Periksa komponen seperti relay, kontaktor dan *power supply* berfungsi baik.

Checking the components such as relays, contactors and the power supply is working normally.



Gambar 11. *Electrical Panel for Wiring and Connection*

Figure 11. Electrical Panel for Wiring and Connection

7.1.9. Pemeriksaan Pompa Vakum. Sebelum memeriksa pompa vakum, pastikan mesin dalam kondisi off. Pompa vakum mesin *Filling* berada di *mezzanine* 1 di atas ruang *Filling Tofflon.* Terdapat 1 Pompa vakum untuk *stoppering.* Pastikan oli terisi, periksa

pada *sight glass* di pompa vakum. Pastikan di tengah-tengah, tidak kurang dan lebih lihat gambar 12. Gunakan oli *vacuum pump* untuk menambah oli jika oli kurang.

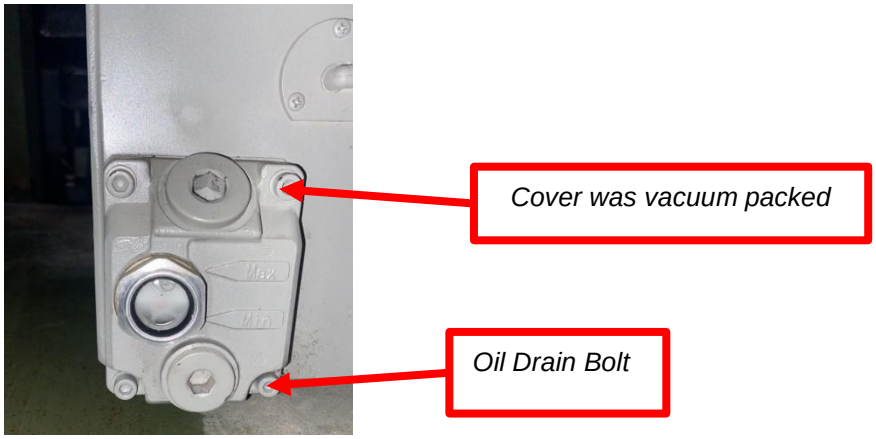
Vacuum Pump Inspection. Before checking the vacuum pump, make sure the machine is off. The Filling machine vacuum pump is located in mezzanine 1 above the Tofflon Filling room. There is 1 vacuum pump for stoppering. Make sure the oil is filled, check the sight glass on the vacuum pump. Make sure it is in the middle, not less and more see figure 12. Use the vacuum pump oil to add oil if the oil is less

Gambar 12. *The Sight Glass for Vacuum Pump Oil*

Figure 12. The Sight Glass for Vacuum Pump Oil

7.1.10. Pemeriksaan Oli Pompa Vakum

Checking the Vacuum Pump Oil



Gambar 13. Pompa Vakum Stoppering

Figure 13. Stoppering Vacuum Pump

7.1.10.1. Cara melakukan penggantian atau memasukan oli *vacuum.*

a. Buka *cover* oli masuk.

   Open the cover of oil in.

b. Siapkan wadah bekas oli di bawah bukaan oli keluar.

Preparing the tub or jerrcan near the hole of out the oil.

c. Buka baut oli keluar.

Open the bolt of oil out.

d. Tunggu sampai tidak ada lagi oli yang mengalir dari pompa vakum.

Wait until no more oil flows from the vacuum pump.

e. Tunggu baut oli keluar, pastikan rapat. Bila perlu gunakan *seal*/tahan panas.

Close the bolt of oil out, tightening the bolt. If needed using the heat resistanst seal.

f. Isi kembali dengan oli baru.

Refill with a new oil.

g. Tutup *cover* oli masuk.

Close the cover of oil in.

7.1.11. Penggantian *Filter* Oli Pompa Vakum

Changing the Filter of Vacuum Pump Oli

7.1.11.1 *Filter* oli *vacuum pump* berada di atas dari indikator oli.

Location the filter at top side from the oil indicator

7.1.11.2 Lepas *Cover filter*. Buka dari kuncian seperti yang ditunjukkan pada gambar 14.

Open the filter cover. Unlock the key like in the figure 14.

Gambar 14. Penutup Filter Pompa Vakum

Figure 14. Vacuum Pump Filter Cover.

7.1.11.3 Ambil filter oli, dan ganti dengan yang baru.

Changing the filter with a new one.

Gambar 15. Filter Pompa Vakum

Figure 15. Vacuum Pump Filter

7.1.12. Pemeriksaan *shaft stoppering*

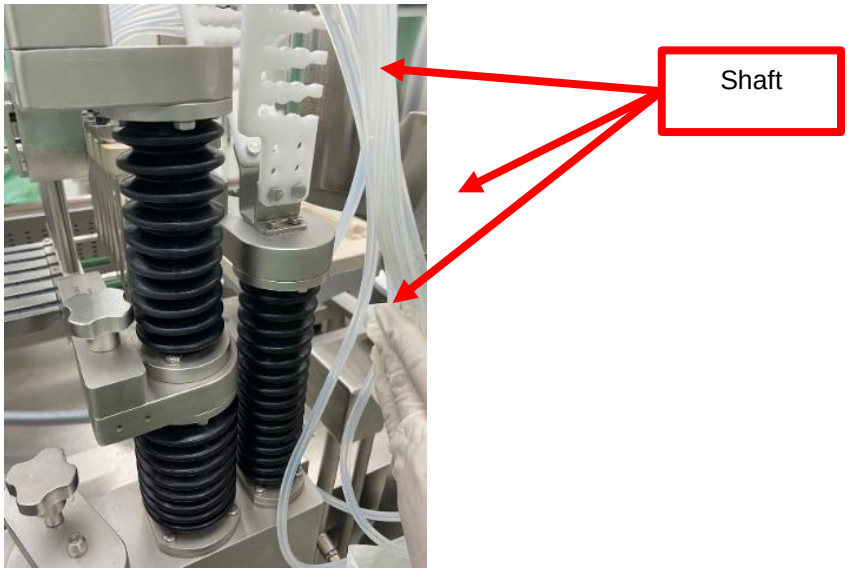
Checking of shaft stoppering

Periksa *shaft,* seperti :

Checking the shaft, such as :

7.1.12.1 Periksa kondisi *Shaft,* pastikan tidak ada sobek pada bagian tersebut

Checking the shaft condition,make sure there are no tears in the part.



Gambar 16. *Shaft stoppering*

Figure 16. Shaft stoppering

7.1.13. Pemeriksaan kondisi *Vibrator Bowl*

Checking Vibrator Bowl Condition

7.1.13.1 Pemeriksaan kondisi *vibrator bowl* pastikan getaran pada *bowl* normal sehingga *rubber* yang ditempatkan pada *bowl* bergerak dengan sempurna

Make sure the vibration in the vibrator bowl is normal to ensure that the rubber placed in the bowl moves perfectly by checking its condition.

Gambar 17. Bowl

Figure 17. Bowl

7.1.14. Pemeriksaan pada *Linear Track*

Checking the Linear Track

7.1.14.1 Pemeriksaan kondisi *linear track* pastikan getaran pada *track* normal sehingga *rubber* yang berada pada track bergerak dengan sempurna

Gambar 18. *Linear Track*

Figure 18. Linear Track

7.1.15. Pemeriksaan pada *Vacuum Regulator Arm Robot*

Vacuum Regulator Arm Robot Check

7.1.15.1 Pemeriksaan vacuum regulator arm robot, pastikan *value* sudah sesuai.

Check the function of the vacuum regulator arm robot, make sure the value is correct.

Gambar 19. Vacuum Regulator Arm Robot

Figure 19. Vacuum Regulator Arm Robot

8. REFERENSI

REFERENCE

Petunjuk Pengoperasian dan Pemeliharaan Mesin *Filling Tofflon.*

Operation and Maintenance Instructions of Filling machine Tofflon

9. LAMPIRAN

ATTACHMENT

Formulir No. SOP/EBI/EN-044-F01B perawatan berkala terhadap Mesin *Filling Tofflon*.

Form No. SOP/EBI/EN-044-F01B for periodic maintenance of the Tofflon Filling Machine.

10. RIWAYAT

HISTORY

Tanggal ***Date***	**No. Change Control** ***Change Control No.***	**No. Revisi** ***No. Revision***	**Alasan Perubahan** ***Reason of Change***
26 Sep 2023	CC-EBI-EN-017-24	00	Dokumen baru *New Document*

	CC-EBI-EN-013-25	01	Menambahkan poin pemeriksaan Kondisi & Fungsi *Regulator Vacuum Robot Arm*, pada poin 7.1.1. *Added Vacuum Robot Arm Regulator Condition & Function check point, at point 7.1.1.*

11. DOKUMEN TERKAIT

RELATED DOCUMENT

Tidak Ada

N/A

12. TARGET PELATIHAN

TRAINING TARGET

Departemen ***Department***	**Jabatan Personil** ***Personil Title***
Departement Engineering *Engineering Departement*	Teknisi *Technician*

13. DISTRIBUSI

DISTRIBUTION

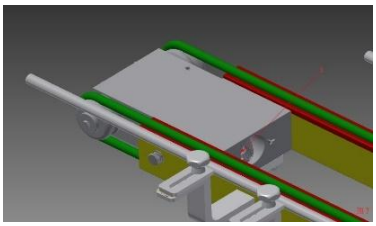
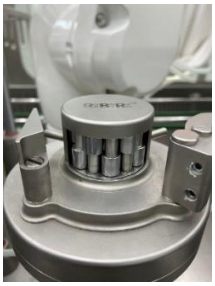
13.1. Departemen Engineering *(Hard Copy).*
Engineering Department (Hard Copy).

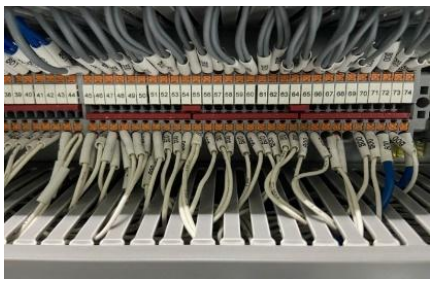
13.2. Departemen Produksi *(Hard Copy).*
Production Department (Hard Copy).

13.3. Departemen Sistem Mutu *(Hard Copy).*
Quality System Departement (Hard Copy).

PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / *FORM* **PERAWATAN BERKALA TERHADAP MESIN FILLING TOFFLON** ***PERIODIC MAINTENANCE OF FILLING MACHINE TOFFLON***	Document No. : SOP/EBI/EN-044-F01B

Level	:
Month	:
Date	:
Day	:
Hours Meter	:

Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L2											
	Lubrication of the filling machine										
											
	Sensor and Detector Check										
	Gear inspection										
	Check on the Peristaltic Pump										
											

Level	Component	Activity							Condition		Remarks
		C	I	L	R	S	T	W	B	G	
	Stoppering rubber condition check										
	Conveyor belt inspection										
	Connection Check on the Electrical Panel										
											
	Vacuum pump inspection										

PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / *FORM* **PERAWATAN BERKALA TERHADAP MESIN FILLING TOFFLON** ***PERIODIC MAINTENANCE OF FILLING MACHINE TOFFLON***	Document No. : SOP/EBI/EN-044-F01B

Level	Component	Activity							Condition		Remarks
		C	I	L	R	S	T	W	B	G	
	Vacuum Pump Filter Check										
	Vibrator Bowl Condition Check										
	Checks on Linear Track										
	Shaft stoppering check										

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Level	Component	Activity							Condition		Remarks
		C	I	L	R	S	T	W	B	G	
	Check Condition & Function of Robot Arm Vacuum Regulator										

Note :

No	Date	Start	Finish	Duration		Total personnel	Area	Remark
				Hour	Minute			

Remark :	Level :	Done by :	Checked by :
C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek)	L1 : Every 1 month	( )	( )